BYTECUBIT TECHNOLOGIES · INDUSTRIAL AUTOMATION ENGINEER

Russell Dudek

Industrial Systems Modernization | Manufacturing Data & Automation

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PROFILE

Operator-technologist with verified depth across high-reliability manufacturing, engineering, production systems, SQL/data workflows, quality, telemetry, and software-enabled operations. Builds mechanisms that make technical work visible, testable, supportable, and connected to frontline outcomes.

~1,000
fielded robots informing current support and operating feedback loops

~5 million
second-day orders annually supported by Amazon Prime Pittsburgh

100%
safety record across regulated drone operations with no FAA incidents

DoD · CERN · Intel · AMD
high-consequence technical collaboration in advanced electronics

ROLE-ALIGNED CAPABILITIES

Manufacturing systems modernization	**SQL and operational data workflows**	**MES / ERP workflow integration**
Industrial automation translation	**Quality systems and root cause**	**Cutover, rollback and standard work**
Telemetry and machine-fleet signals	**Cross-functional technical delivery**	**AI/agentic workflow design**

SELECTED EXPERIENCE

Vape-Jet — 2025-Present

Director of Operations

- Leads AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows.
- Connects factory, field, customer, and machine-fleet signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
- Works directly in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots informing support and operating feedback loops.
- Builds visible issue, quality, and workflow mechanisms across technical and frontline teams rather than relying on person-dependent knowledge.

Compunetics — 2000-2013

Director of Operations and Engineering Management

- Progressed through R&D program management, production management, strategic acquisition, operations management, and director-level leadership.
- Led engineering, manufacturing, quality, customer delivery, and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, standard work, root-cause practices, performance measures, and process controls across technical operations.
- Collaborated with high-consequence customers and partners including DoD, CERN, Intel, and AMD.
- Published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.

Amazon — 2014-2018

Operations Management / Site Leader

- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually.
- Built operating routines, labor planning, throughput visibility, escalation paths, standard work, and performance mechanisms.
- Applied Lean and Gemba practices across safety, quality, throughput, customer experience, and team engagement.
- Served on the Amazon Prime Now launch team with Whole Foods and contributed to Prime Air research and Amazon Flex geolocation concepts.

SYSTEMS THROUGH-LINE

> Across robotics manufacturing, advanced electronics, and continuous operations, I have repeatedly connected technical change to production behavior, quality, data visibility, ownership, support, and customer consequence.

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ADDITIONAL EXPERIENCE

DudeWorth
2017-Present

Founder / AI and Automation Advisor

- Develops readiness and opportunity frameworks that assess value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value.
- Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails.
- Advises on operations, logistics technology, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis, and workflow design.

Cardinal Building Products
2021-2023

Regional Director

- Built the company's first remote regional operation and modernized distributed customer, CRM, fulfillment, logistics, digital marketing, and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts.

ZeusVu
2018-2022

Chief Pilot, Autonomous Systems and Aerial Data Operations

- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.

INDUSTRIAL DELIVERY METHODS

Current-state and dependency mapping	**Function and exception decomposition**	**SQL/data lineage and context review**
Quality and failure-mode analysis	**Cutover, rollback and observation**	**Diagnostics, runbooks and ownership**
Standard work and operator adoption	**Reusable workflow/component patterns**	**Human-authority AI governance**

TECHNICAL TOOLKIT

- SQL
- Python
- JavaScript
- TensorFlow
- Tableau
- ERP/Odoo workflows
- Jira / Confluence
- HubSpot / Slack
- Agentic AI / LLM workflows
- RAG patterns
- Human-in-the-loop design
- GIS / mapping workflows

EDUCATION AND CREDENTIALS

University of Pittsburgh
Bachelor of Science
Academic grounding includes materials science and physical chemistry.
Google Scholar profile

Credentials
Google AI Essentials · IBM Enterprise Design Thinking Practitioner · Six Sigma Certification · OSHA 10

TECHNICAL LEADERSHIP

IEEE - CMPT and EDS Pittsburgh Chapter Chair, 2012-2016: Led programming connecting industry, academia, electronics, manufacturing, STEM outreach, and emerging technology.

CIVIC LEADERSHIP

Enviro Social Capital - Chairman of the Board, 2014-2017: Co-founded a public-private impact initiative focused on sustainable infrastructure, green finance, and civic transformation.

IMMEDIATE CONTRIBUTION

> Manufacturing-system discovery, SQL and data-context mapping, quality and release discipline, cross-functional technical execution, operator/support adoption, and rapid contribution through disciplined platform learning.